

Lecture 5

Incomplete Information

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States

Key formalism for games of incomplete information is the notion of a **state**

Each state is a complete description of one possible way the world is

- Preferences over outcomes associated with profiles of actions
- (Information about other players' information—not today)

At the start of the game, a state is realized. The state θ is drawn from a **state space** Θ

Signals

Decision Maker (DM) does not directly observe the state

Instead, observes a **signal** that may give her information about the state

In state $\theta \in \Theta$, player i receives signal $\tau_i(\theta)$

The function $\tau_i : \Theta \rightarrow T_i$ is i 's **signal function**. Here, we assume that it is deterministic (for each element of Θ , only one element of T_i ; probabilistic case is harder)

States that generate signal t_i are **consistent with t_i** :

$$\tau_i^{-1}(t_i) = \{\theta \mid \tau_i(\theta) = t_i\}$$

Learning from Signals

The size of the sets of states consistent with each signal reflects quality of i 's information

If only state θ' generates signal t' , then observing t' fully reveals the state

If τ_i is the same for all θ , then signal completely uninformative

Examples

The trivial (uninformative) signal:

$$\tau_i(\theta) = t \text{ for all } \theta \in \Theta$$

Fully revealing signals:

$$\theta \neq \theta' \Rightarrow \tau_i(\theta) \neq \tau_i(\theta')$$

A partially revealing signal: take $\Theta = \{1, 2, 3\}$ and $T_i = \{a, b\}$, and let

$$\tau_i(1) = \tau_i(2) = a \text{ and } \tau_i(3) = b$$

Then $\tau_i^{-1}(a) = \{1, 2\}$ and $\tau_i^{-1}(b) = \{3\}$

Beliefs

We adopt the standard Bayesian approach

Before getting her signal, DM has beliefs about the state represented by a prior probability distribution, $p : 2^\Theta \rightarrow [0, 1]$

After receiving the signal t_i , DM updates to the conditional distribution

$$p(\theta \mid t_i) = \frac{p(\theta)}{\sum_{\theta' \in \tau_i^{-1}(t_i)} p(\theta')}$$

Fact This formula is a consequence of Bayes' rule

A non-deterministic signal

Three states $\Theta = \{D, T, R\}$, two signal realizations
 $T = \{d, n\}$

Prior: $p(D) = 2/6, p(T) = 1/6, p(R) = 3/6$

Three distributions:

$$\Pr(\tau = d \mid D) = 5/6$$

$$\Pr(\tau = d \mid T) = 3/6$$

$$\Pr(\tau = d \mid R) = 1/4$$

Exercise: Compute $\Pr(\theta = D|d), \Pr(\theta = T|d), \Pr(\theta = R|d)$. Compare with prior.

Decision Problems

After getting her signal, DM will choose an action from A

Let $u : A \times \Theta \rightarrow \mathbb{R}$ be her utility function

Two ways to approach the problem

- Interim decision problem: given t_i ,

$$\max_{a \in A} \sum_{\theta} u(a, \theta) p(\theta \mid t_i)$$

- Ex-ante decision problem: choose a function $a^* : T_i \rightarrow A$ to solve

$$\max_{a(\cdot)} \sum_{\theta} u(a(\tau(\theta)), \theta) p(\theta)$$

Fact: These approaches give the same answer

Comparison of Information Structures

It seems intuitively obvious that having more information helps a decision maker. Can we formalize this?

Let $\tau : \Theta \rightarrow T$ and $\sigma : \theta \rightarrow S$ be two signal functions

Say that τ is **at least as informative as** σ if

$$\tau(\theta) = \tau(\theta') \text{ implies } \sigma(\theta) = \sigma(\theta')$$

Implies set of states consistent with some s_i is the union of sets of states consistent with t_i for some collection of t_i

Examples

Let $\Theta = \{1, 2, 3, 4\}$ and let $\sigma(1) = \sigma(2) = a$ and $\sigma(3) = \sigma(4) = b$

If $\tau(1) = a$, $\tau(2) = b$, and $\tau(3) = \tau(4) = c$, then τ is at least as informative as σ

If $\tau(1) = a$ and $\tau(2) = \tau(3) = \tau(4) = b$, then neither τ or σ is at least as informative as the other

DM endowed with σ would be willing to pay for τ in the first case

More Information Can't Hurt

Proposition

Proposition 1. *Let τ and σ be two signal functions, with τ at least as informative as σ . Then, in a one-person decision problem, the DM's optimal payoff is at least as great with τ as with σ .*

Proof.

Proof.

- Let a^* be the optimal strategy when the signal function is σ
- Since τ is at least as informative as σ , we can replicate strategy a^* when the signal function is τ
- Let $\{t_i^1, \dots, t_i^k\}$ be the τ -signals that come from states that σ maps to s_i
- Choose action $a^*(s_i)$ if signal is t_i^j for some j
- Since this is a feasible strategy under τ , a^* has to be weakly better for DM

□

For a generalized version of this idea: Blackwell (1951, 1953)

Modeling Incomplete Information

A **Bayesian Game** is a strategic form game of incomplete information

Each state is a complete description of one collection of the players' relevant characteristics:

- Preferences over outcomes associated with profiles of actions
- Information (e.g., about other players' characteristics)
- In state $\theta \in \Theta$, player i observes the realization t_i of a signal $\tau_i(\theta)$

An Example

Pl. A unsure about whether Pl. B likes spending time with her or not; she can either go to a concert (C), or to an improv show (I)

Pl. A believes that with probability $1/2$ she is playing:

		Pl. B	
		<i>C</i>	<i>I</i>
Pl. A	<i>C</i>	2, 1	0, 0
	<i>I</i>	0, 0	1, 2

and with complementary probability she is playing:

		Pl. B	
		<i>C</i>	<i>I</i>
Pl. A	<i>C</i>	2, 0	0, 2
	<i>I</i>	0, 1	1, 0

States of the World

We can think of this game as representing a situation with two states:

1. Pl. B likes hanging out with Pl. A
2. Pl. B doesn't like hanging out with Pl. A

Pl. B knows state; Pl. A only has beliefs

Nash Equilibrium should be a profile of actions such that all players are playing best responses **given beliefs about other's actions**, and beliefs should be correct

type Signals and Types

- Your type is just your posterior beliefs about θ

Example

- Two states: B likes A ($\theta = B_L$), B dislikes A ($\theta = B_D$)
- Only one type of Pl. A (i.e., A doesn't know θ):
$$\tau_A(B_L) = \tau_A(B_D)$$
- Two types of Pl. B (i.e., B knows θ): $\tau_B(B_L) = t_L$ and
$$\tau_B(B_D) = t_D$$

Each player may care about both actions taken and payoff-relevant state

- We need to define payoff functions over pairs (a, θ)

Bayesian Games

Definition

A **Bayesian Game** consists of

- a set of players
- a set of payoff-relevant states Θ
- a common prior $p(\theta)$ over Θ

and for each player

- a set of actions A_i
- a set of signals T_i that she may receive and a signal function $\tau_i : \Theta \rightarrow T_i$ that associates a signal to each state
- a payoff function $\pi : A \times \Theta \rightarrow \mathbb{R}$, the expected value of which represents i 's preferences about lotteries of (a, θ)

Payoffs in a Bayesian Game

Denote by $\Pr(\theta|t_i)$ player i 's belief about the probability that the payoff-relevant state is θ given signal t_i

Denote the action by a type t_j of a player j by $a(j, t_j)$

Player j 's signal in state θ is $\tau_j(\theta)$, so her action is $a(j, \tau_j(\theta))$

For each state θ , let j 's action be $\hat{a}_j(\theta) = a(j, \tau_j(\theta))$

The expected payoff of type t_i of player i if she chooses action a_i is

$$\sum_{\theta \in \Theta} \Pr(\theta|t_i) \pi_i((a_i, \hat{a}_{-i}(\theta)), \theta)$$

Nash Equilibrium in a Bayesian Game

Definition

A **NE of a Bayesian Game** is a NE of the strategic form game defined as follows:

- **Players:** The set of all pairs (i, t_i) in which i is a player in the Bayesian game and t_i is one of the signals that i may receive
- **Actions:** The set of actions of each player (i, t_i) is the set of actions of player i in the Bayesian Game
- **Payoffs:** payoff function of each player (i, t_i) is given by

$$\sum_{\theta \in \Theta} \Pr(\theta | t_i) \pi_i(a_i, \hat{a}_{-i}(\theta), \theta)$$

Understanding MSNE through Bayesian Equilibria

Consider the following game G

		Firm 2	
		Invest	Don't Invest
Firm 1	Invest	$-1, -1$	$1, 0$
	Don't Invest	$0, 1$	$0, 0$

3 NE:

- (I, D)
- (D, I)
- $(\sigma_1(I) = \sigma_2(I) = 1/2)$

Turning the Example a Bayesian Game

Consider same game except if a firm is sole investor, gets $1 + \theta_i$

The θ 's are independent and uniformly distributed on $[-\epsilon, \epsilon]$

We refer to this as the **perturbed game** G_ϵ

The (payoff-relevant) state is $\theta = (\theta_1, \theta_2)$

Each firm learns its own θ_i but not others ($\tau_i(\theta_1, \theta_2) = \theta_i$)

Equilibrium of the Bayesian game

Symmetric pure strategy profile “invest if and only if $\theta_i > 0$ ” is a Bayesian Equilibrium of this game

Each firm invests with probability 1/2 from other firm's perspective

Payoff of investing if is

$$-\frac{1}{2} + \frac{1}{2}(1 + \theta_i) = \frac{\theta_i}{2}$$

As ϵ converges to 0, the pure strategy Bayesian equilibrium of G_ϵ converges to the MSNE of the G (with respect to the distribution of strategies)

The Purification Theorem (technical)

Consider a finite strategic form G with action spaces A_i . A (possibly mixed) strategy profile σ^* NE of G is *regular* if the Jacobian determinant of a continuously differentiable map characterizing it is non-zero at σ^*

Theorem

Suppose that σ^* is a regular NE of G ; for all independent, twice-differentiable distributions p_i on $\Theta_i \in [-1, 1]^{\#A}$, σ^* is the limit as $\epsilon \rightarrow 0$ of a sequence of pure-strategy equilibria of the perturbed game G_ϵ with $\tilde{\pi}_i = \pi_i(a) + \epsilon \theta_i^a$; G_ϵ has a pure strategy equilibrium σ_ϵ^* whose induced probability distribution over strategies converges to the distribution induced by σ^*

Does Information Always Help?

Suppose both players believe each state is equally likely

		2		
		L	M	R
1	T	1, 2€	1, 0	1, 3€
	B	2, 2	0, 0	0, 3

STATE θ_1

		2		
		L	M	R
1	T	1, 2€	1, 3€	1, 0
	B	2, 2	0, 3	0, 0

STATE θ_2

States and Types in this Game

Assume $0 < \epsilon < \frac{1}{2}$

There are two states and player 2's payoffs are different in each state

Neither player knows the true state

Each player has only 1 type

Nash Equilibrium

Player 2's unique best response is L to all actions by 1

- $U_2(T, L) = 2\epsilon > U_2(T, M) = U_2(T, R) = \frac{3}{2}\epsilon$
- $U_2(B, L) = 2 > U_2(B, M) = U_2(B, R) = \frac{3}{2}$

Player 1's unique best response to L is B

Unique NE (B, L) yielding 2 to each player

A Variant Where Player 2 is Informed

Suppose Player 2 learns the true payoff relevant state:

$$\tau_2(\theta_1) \neq \tau_2(\theta_2)$$

Player 2 of type $\tau_2(\theta_1)$ has a dominant action: R

Player 2 of type $\tau_2(\theta_2)$ has a dominant action: M

Player 1's unique best response to R or M is T

Unique pure strategy NE: $(T, (R, M))$

Both players are worse off (2's payoff now is $3\epsilon < 2$, 1's is $1 < 2$)

Public Good Provision Game

Two players, 1 and 2, simultaneously decide whether to contribute to a common project (public good)

Public good provided as long as at least one player contributes

Each player derives a benefit of 1 if the good is provided and 0 if it is not

Player i 's cost to contributing is c_i

Exercise Suppose $(c_1, c_2) \in (0, 1)^2$ are common knowledge; what is the set of Nash Equilibria?

Uncertainty over Costs

Each individual knows her own cost of contributing

Each individual believes other player's cost is drawn from a cdf $F(\cdot)$ on $[\underline{c}, \bar{c}]$, where $\underline{c} < 1 < \bar{c}$

A (payoff-relevant) state is a pair (c_1, c_2)

Each player receives a signal $\tau_i(c_1, c_2) = c_i$

A player's cost is his type

Equilibrium Defined

A pure strategy in this game is a function $s_i(c_i) : [\underline{c}, \bar{c}] \rightarrow \{0, 1\}$, where 1 means contribute

Player i 's payoff is:

$$\pi_i(s_i, s_j, c_i) = \max(s_1, s_2) - c_i s_i$$

A (Nash) Bayesian equilibrium is a strategy profile $(s_1^*(\cdot), s_2^*(\cdot))$ such that for each player i and every possible value of c_i :

$$s_i^*(c_i) \in \arg \max_{s_i} \mathbb{E}_{c_{-i}} [\pi_i(s_i, s_{-i}^*(c_{-i}), c_i)] \iff$$

$$s_i^*(c_i) \in \arg \max_{s_i} \int_{\underline{c}}^{\bar{c}} \pi_i(s_i, s_{-i}^*(\tilde{c}), c_i) f(\tilde{c}) d\tilde{c}$$

Solving for Equilibrium

Let q_i be the equilibrium probability that a player i contributes

Player i 's payoff from contributing is $1 - c_i$

Player i 's payoff from not contributing is q_{-i}

Player i contributes if $1 - c_i \geq q_{-i}$ or $c_i \leq 1 - q_{-i}$

Call this critical value $c_i^* \equiv 1 - q_{-i}$

Player i contributes only if $c_i \in [\underline{c}, c_i^*]$, where $c_i^* = 1 - q_{-i}$

In equilibrium, the probability that player $-i$ contributes is $q_{-i} = F(c_{-i}^*)$

Equilibrium II

This implies that equilibrium is a pair (c_1^*, c_2^*) that solves

$$c_1^* = 1 - F(c_2^*)$$

$$c_2^* = 1 - F(c_1^*)$$

A symmetric eqm ($c^* : 1 - F(c^*) - c^* = 0$) always exists. Let

$$\mathcal{F}(c) = 1 - F(c) - c$$

Note $\mathcal{F}(\underline{c}) = 1 - \underline{c} > 0$, $\mathcal{F}(\bar{c}) = -\bar{c} < 0$, $\mathcal{F}' = -f(c) - 1 < 0$.

Exercise Solve for c^* under the assumption that $F(\cdot)$ is uniformly distributed in $[\gamma - \frac{1}{2\psi}, \gamma + \frac{1}{2\psi}]$

Equilibrium III

If there is a unique c^* that solves $c^* = 1 - F(1 - F(c^*))$, then necessarily $c_i^* = c^* = 1 - F(c^*)$

For instance, if $\max f(c) < 1$, we have that the mapping

$$\mathcal{G}(c) = 1 - F(1 - F(c)) - c$$

satisfies $\mathcal{G}'(c) < 0$, $\mathcal{G}(\bar{c}) < 0$, $\mathcal{G}(\underline{c}) > 0$ (can you prove the latter?)

But in general there may be multiple non-symmetric equilibria, depending on the shape of $F(\cdot)$

Multiple Players

Consider the same game with n players; we will look for a symmetric NE (all players contribute with probability p)

Probability at least one person other than you contributes is $1 - (1 - p)^{n-1}$

A player will contribute if $1 - c_i \geq 1 - (1 - p)^{n-1}$ or $c_i \leq (1 - p)^{n-1} \equiv c^*$

At symmetric equilibrium, $c^* = (1 - p)^{n-1}$, i.e., the solution of

$$\mathcal{F}_n(c) = (1 - F(c))^{n-1} - c = 0$$

As n increases, c^* must decrease (can you prove it?)

Does it imply lower public good provision? No! (Can you prove the opposite?)

A Model of Juries:

Each juror comes to the trial with believing that the defendant is guilty with probability π

From the evidence, each juror gets one of two impressions, guilty (g) or innocent (i)

Juror gets impression guilty with probability p , where $p > 1/2$ if and only if the defendant is actually guilty

Juror gets the impression innocent with probability q , where $q > 1/2$ if and only if defendant is innocent

Jurors suffer a disutility $-z$ if convict innocent and $-(1 - z)$ if acquit guilty

Unanimity is required for conviction

What the paper does

Study of Unanimity vote in Juries

Unanimity established as a uniquely bad due to strategic voting

- In any equilibrium under unanimity in which the defendant is convicted with positive probability, the probability of convicting an innocent is bounded below by a strictly positive number
- **Under symmetric prior**, for any non-unanimous rule there exists an equilibrium in which the probability of mistakes converges to 0 for n large enough

Expected Payoffs

Player i 's payoffs are:

$$\pi_i(a, \theta) = \begin{cases} -z & \text{if } a = (C, \dots, C) \text{ and } \theta = I \\ -(1 - z) & \text{if } a \neq (C, \dots, C) \text{ and } \theta = G \\ 0 & \text{otherwise} \end{cases}$$

where θ is the defendant's true status (payoff-relevant state)

Defining the Bayesian Game

Players the set of n jurors

Payoff-relevant state $\theta \in \{G, I\} \equiv \Theta$

Prior $\pi = \Pr(\theta = G)$

Actions The set of actions for each player is $\{C, Q\}$ where C means vote for conviction and Q for acquitting

Signal (functions and spaces) The set of signals that each player may receive is $\{g, i\}$ and player j 's signal function is defined by $\tau_j(\theta) = s_j$ (jurors only knows her own impression)

[Some textbooks define State Spaces more broadly to include signal realizations: $\Omega \equiv \Theta \times \{g, i\}^n$]

Defining Bayesian Game (cont'd.)

Beliefs, contingent on all signals are updated using Bayes' Rule

$$P(A|B)P(B) = P(A \cap B) = P(B|A)P(A)$$

Payoff functions: expected utilities given priors and beliefs (as above)

Let's start with one juror

If there is one juror who receives signal i her beliefs are

$$\Pr(G|i) = \frac{\Pr(i|G) \Pr(G)}{\Pr(i|G) \Pr(G) + \Pr(i|I) \Pr(I)} = \frac{(1-p)}{(1-p)\pi + q}$$

Payoff from acquitting:

$$-\frac{(1-p)\pi}{(1-p)\pi + q(1-\pi)}(1-z)$$

Payoff from convicting:

$$-\left(1 - \frac{(1-p)\pi}{(1-p)\pi + q(1-\pi)}\right)z$$

One Juror (con'd)

Acquit following i if more concerned about convicting an innocent:

$$z \geq \frac{(1-p)\pi}{(1-p)\pi + q(1-\pi)} = \left[1 + \frac{q}{1-p} \frac{1-\pi}{\pi} \right]^{-1}$$

If receives signal g , acquit iff:

$$z \geq \frac{p\pi}{p\pi + (1-q)(1-\pi)} = \left[1 + \frac{1-q}{p} \frac{1-\pi}{\pi} \right]^{-1}$$

More likely to acquit following a signal of innocent

Follow her signal own signal iff balance in concerns:

$$\frac{(1-p)\pi}{(1-p)\pi + q(1-\pi)} \leq z \leq \frac{p\pi}{p\pi + (1-q)(1-\pi)}$$

Two jurors

Are there values of z such that the game has a NE in which each juror votes according to her signal?

Suppose juror 2 acts in this way: type i votes to acquit and type g votes to convict

Consider type i of juror 1

- If juror 2's signal is i , juror 1's vote has no effect on outcome
- When deciding how to vote, juror 1 ignores the cases where juror 2's signal is i

Juror 1's Decision Problem

$$u_1(Q|s_1 = i) = \Pr(s_2 = i) [\Pr(I|i, i) \times 0 - \Pr(G|i, i) \times (1 - z)] \\ + \Pr(s_2 = g) [\Pr(I|i, g) \times 0 - \Pr(G|i, g) \times (1 - z)]$$

$$u_1(C|s_1 = i) = \Pr(s_2 = i) [\Pr(I|i, i) \times 0 - \Pr(G|i, i) \times (1 - z)] \\ + \Pr(s_2 = g) [-\Pr(I|i, g) \times z + \Pr(G|i, g) \times 0]$$

Follow signal (acquit) if

$$\Pr(s_2 = g) (-\Pr(G|i, g)(1 - z) + \Pr(I|i, g)z) \geq 0$$

This is equivalent to $\Pr(G|i, g) \leq z$

Will Juror 1 Follow His Signal?

Type i of juror 1 follows signal if $\Pr(G|i, g) \leq z$

$$\Pr(G|i, g) = \frac{(1-p)p\pi}{(1-p)p\pi + q(1-q)(1-\pi)}.$$

Type i of juror 1 votes to acquit iff

$$z \geq \frac{(1-p)p\pi}{(1-p)p\pi + q(1-q)(1-\pi)}.$$

By a similar argument, type g of juror 1 votes to convict iff

$$z \leq \frac{p^2\pi}{p^2\pi + (1-q)^2(1-\pi)}$$

Comparing 1 and 2 Jurors

Game with two jurors has NE in which jurors act according to signals iff:

$$\frac{(1-p)p\pi}{(1-p)p\pi + q(1-q)(1-\pi)} \leq z \leq \frac{p^2\pi}{p^2\pi + (1-q)^2(1-\pi)}$$

Given $p > 1 - q$,

$$\frac{(1-p)p\pi}{(1-p)p\pi + q(1-q)(1-\pi)} > \frac{(1-p)\pi}{(1-p)\pi + q(1-\pi)}$$

Lowest z for which equilibrium exists in which all jurors vote according to signal higher with two jurors

A type i juror, knowing vote only matters if other juror votes C , votes **as if** other juror's signal is g , and so is less worried about convicting innocent than without other juror

Many jurors

Suppose the number of jurors is n

Suppose that every juror other than juror 1 votes according to her signal

Juror 1's vote has no effect on the outcome unless every other juror's signal is g

Let $g_{-1} \equiv (g, \dots, g)$

When deciding how to vote after receiving a signal of i , the juror considers

$$\begin{aligned}\Pr(G|i, g_{-1}) &= \frac{\Pr(i, g_{-1}|G) \Pr(G)}{\Pr(i, g_{-1}|G) \Pr(G) + \Pr(i, g_{-1}|I) \Pr(I)} \\ &= \frac{(1-p)p^{n-1}\pi}{(1-p)p^{n-1}\pi + q(1-q)^{n-1}(1-\pi)}\end{aligned}$$

Is Unanimity Rule for Conviction Bad for the Innocent

Type i juror 1 optimally votes for acquittal iff (after rearranging)

$$z \geq \left[1 + \frac{q}{1-p} \left(\frac{1-q}{p} \right)^{n-1} \frac{1-\pi}{\pi} \right]$$

Given $p > 1 - q$, lower bound on z for which type i Juror 1 will vote to acquit approaches 1 as n increases

In a large jury, if jurors care at all about acquitting the guilty, juror who think innocent will nevertheless vote convict because only pivotal if everyone else thought guilty

In large jury, no NE in which every juror votes according to her signal unless don't care at all about acquitting the guilty

Mutual Optimism as a Cause for War:

General idea: if two countries agree on the probability each wins a war, then war cannot be Pareto optimal

This leaves open the idea that war can be Pareto optimal in a situation of “mutual optimism”—each country believes that it is more likely to win

In his classic paper on “Rationalist Explanations for War”, Fearon makes two points:

- In the standard Bayesian game framework, this requires the countries to have seen different signals
- and this creates incentives for information sharing

Fearon is understating the difficulties of the mutual optimism argument in a Bayesian game: even without communication, mutual optimism cannot lead to war being Pareto optimal

A Simple Model of War

Two countries (1 and 2) can either agree to fight a war or not

Each has two actions: F and N . War is outcome (F, F)

- No unilateral wars—lets us focus on Pareto optimality claim

Choosing N leads to a payoff of 0 for sure

Choosing F incurs mobilization cost μ whether or not the war happens

If the war happens, the winner takes resources worth 1 from the loser

- Actual costs of war strengthen result

Information Structure and Payoffs

Set of states is Ω , signal functions are τ_1 and τ_2

- A state determines who wins and what each country knows

We make no assumptions about the details of the information structure

$$u_i(N, a_j, \omega) = 0 \text{ for all } a_j \in \{F, N\} \text{ and } \omega \in \Omega$$

$$u_i(F, N, \omega) = -\mu \text{ for all } \omega \in \Omega$$

$$u_i(F, F, \omega) \in \{1 - \mu, -1 - \mu\} \text{ for all } \omega \in \Omega$$

$$u_1(F, F, \omega) + u_2(F, F, \omega) = -2\mu \text{ for all } \omega \in \Omega$$

Equilibrium

Exercise Show that (N, N) is a Bayesian equilibrium

Can there be any other equilibrium? No

Proposition

Proposition 2. *There is a unique equilibrium, in which each country chooses N for all signals*

Proof

Assume that there is an equilibrium in which country 1 chooses F for a non-empty subset $\Omega_1^F \subseteq \Omega$

Exercise What does this imply about the other player's behavior?

Implies that $\Omega_F \equiv \Omega_1^F \cap \Omega_2^F$ is non empty, otherwise N strictly dominates this F for 1

Let $x_i(\omega)$ be the gross resource gain for country i at state ω if there was a war

- Notice that x does not depend on action played

Notice that $x_1(\omega) + x_2(\omega) = 0$ for every state ω

Proof (cont'd)

If country 1 plays F for each signal $t \in T_1^F \subset T_1$, we must have

$$\sum_{\omega \in \tau^{-1}(t)} p(\omega | t) x_1(\omega) > 0 \quad \forall t \in T_1(F)$$

Hence,

$$\sum_{t \in T_1^F} p(t) \sum_{\omega \in \tau^{-1}(t)} p(\omega | t) x_1(\omega) = \sum_{\omega \in \Omega_1^F} p(\omega) x_1(\omega) > 0$$

Notice that

$$\sum_{\omega \in \Omega_1^F} p(\omega) x_1(\omega) = \sum_{\omega \in \Omega_F} p(\omega) x_1(\omega) + \sum_{\omega \in \Omega_1^F / \Omega_F} p(\omega) \times 0$$

Proof (cont'd)

Since i's expected payoff from playing F is given by

$$\sum_{\omega \in \Omega_F} p(\omega) x_i(\omega) \times 1 - \mu$$

we must have $\sum_{\omega \in \Omega_F} p(\omega) x_i(\omega) > 0$ for both players

This implies

$$\sum_{\omega \in \Omega_F} p(\omega) [x_1(\omega) + x_2(\omega)] > 0$$

a contradiction

References